

YASSINE HAJ HMIDA

Industrial Instrumentation & Maintenance Engineering Student · Embedded Systems · IoT · Robotics · Founder

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[Portfolio](#) · [LinkedIn](#) · [GitHub](#)

EDUCATION

National Institute of Applied Sciences and Technology (INSAT) — Tunis 2023 – 2028 (expected)

Engineering Degree — Industrial Instrumentation & Maintenance Engineering

- Completed integrated preparatory cycle (mathematics, physics, computer science); core specialization in instrumentation, maintenance, automation, electronics, data acquisition and embedded systems.

Technical Baccalaureate — Highest Honors — Lycée Pilote de l’Ariana, Tunis 2023

EXPERIENCE & ENTREPRENEURSHIP

Dhawily — Founder & Developer 2024 – Present

- End-to-end management of a digital fabrication initiative (custom LED neon signs, CNC machining, acrylic engraving) — design, production, marketing and customer relations — delivering to real paying clients.
- Built an AI pipeline that converts text and images into production-ready CNC files, respecting real Neon Flex constraints (bend radius, segment continuity, channel width) to cut design time and material waste.

SELECTED ENGINEERING PROJECTS

Smart Air Quality Monitor IoT & Embedded Systems

ESP32-based system that measures air-quality indicators and streams them in real time to a live dashboard.

Stack: ESP32 · sensors · Wi-Fi · IoT

Haniny — Smart Driving & Insurance Hackathon · Mobile · Top 4 / 50+ teams

Mobile app that scores driving risk for usage-based insurance, built under hackathon pressure.

Stack: Flutter · risk analysis · UX

Arduino Robotic Arm Robotics & Mechatronics

Multi-axis servo-actuated arm with sensor feedback; designed from scratch and featured on national television.

Stack: Arduino · servomotors · sensors · embedded C

Flood Identification Dashboard — Tunisia Data & Front-End

Interactive dashboard mapping flood-risk areas across Tunisia from geographic and risk data.

Stack: data visualization · GIS · front-end

Others Wino (services marketplace, Flutter) · Wassalni (mobility app).

TECHNICAL SKILLS

Programming C · C++ · Python · JavaScript · TypeScript · Dart / Flutter · HTML / CSS

Embedded Systems & IoT Arduino · ESP32 · microcontrollers · sensor integration · data acquisition · real-time monitoring · IoT systems

Industry & Manufacturing instrumentation · maintenance · automation · CNC / CAM · CAD · product design

Software & Tools mobile (Flutter) · front-end & dashboards · Photoshop · Illustrator · After Effects · Premiere Pro

AWARDS & ACHIEVEMENTS

- Baccalaureate ranked 72nd nationally out of 17,249 candidates (17.93/20, Highest Honors) — top 0.5%.
- 4th place out of 50+ teams — JE INSAT Hackathon (Haniny).
- 1st place, Tunisian National Robotics Day (TUNIROBOTS), INSAT, 2018 — autonomous task-performing robot.

LANGUAGES

Arabic (native) · French (professional, B2) · English (professional, B2) · Spanish (intermediate)